

HS 123: Game Theory and Economics
BTech 4th Semester, Quiz 1
Total Marks 10, Time 55 minutes
Date: 7th Feb, 2024
[Answers should be accompanied by proper elaboration]

1. Two political parties, L and R , are fighting an election, x and y (non-negative numbers) are their advertisement expenditures respectively. For L , the share of votes it gets is given by, $\frac{x}{x+y}$. For simplicity, we assume that the payoff to L is given by, $100 \frac{x}{x+y} - x$. Similarly, for R it is, $100 \frac{y}{x+y} - y$.
 - (a) Specify this situation as a strategic game and find the best response functions of the parties. [1+2]
 - (b) What is (are) the Nash equilibrium (equilibria) of the game? [2]
 - (c) Does this game resemble any basic template of games you have seen before? [1]
2. Give an example of a two-player strategic game in which each player has finitely many actions and in the only Nash equilibrium both players' actions are weakly dominated. [4]

HS 123: Game Theory and Economics
BTech, 4th Semester, Mid-Semester Test
Total Marks: 40

Date: 1st March, 2024

Elaboration of answers is a must

1. Which actions are called weakly dominated actions? In the following game, which action profiles survive after iterated elimination of strictly dominated actions? Explain. [2+4]

		B		
		L	M	R
A	T	3,1	2,3	10,2
	H	4,5	3,0	6,4
	L	2,2	5,4	12,3
	B	5,6	4,5	9,7

2. 50 students play the following game. Each chooses an integer from 0 to 100. Let X = the mean of the 50 chosen numbers. The student who has chosen a number which is closest to the two-thirds of X wins a prize of 100 rupees. If multiple students tie, they split the prize equally.
- Find a symmetric Nash equilibrium of the game (all players' actions are same). [3]
 - Is the symmetric Nash equilibrium unique? [3]
 - Suppose all else have chosen 30, then for an individual player i , what is the set of best responses? [6]
3. In a market with Cournot duopoly two firms 1 and 2 sell a product whose the inverse demand function is given by, $P = 30 - Q$, if $Q \leq 30$; and $P = 0$, if $Q > 30$. The cost functions of the firms are, $C_i(q_i) = 2q_i$.
- In the Nash equilibrium what is output level of the firms? [3]
 - In the Nash equilibrium what is the price of the product and how much is the individual profit of the two firms? [3]
 - Suppose, firm 1 discovers a new technology through its R&D. Its cost function is now, $C_1(q_1) = q_1$. Firm 2 has the same cost function as before. What are equilibrium output levels in the changed situation and how much are the profits? Interpret the result by comparing it to your answer in part (a) and (b). [6]
4. Consider a Cournot model with n firms. The demand function is, $P = \alpha - Q$, if $Q \leq \alpha$; and $P = 0$, if $Q > \alpha$. The cost functions are, $C_i(q_i) = cq_i$, $\alpha > c > 0$.
- Derive the best response function of each firm. [2]
 - Assuming that there is a Nash equilibrium where each $q_i^* > 0$, prove that all q_i^* are equal. Find the value of q_i^* . [4]
 - Find the equilibrium price and show that it approaches c as n goes to infinity. [4]

HS 123: Game Theory and Economics
BTech 4th Semester, Quiz 2
Total Marks 10, Time 55 minutes
Date: 3rd April, 2024
[Answers should be accompanied by proper elaboration]

1. [Voting game with vote bank] Recall the setting of the electoral competition game. Suppose the distribution of favourite positions of voters has median at m . There are two candidates, 1 and 2.
 - (i) 20% of the extreme right-wing voters vote *in block* for the candidate whose position is more to the right. In case both candidates take the same position, the votes of the block get evenly distributed. The rest 80% of voters vote individually like in the original setting. Find the Nash equilibrium/equilibria. [2]
 - (ii) The block votes for 1 only, irrespective of the position 1 takes. Identify any one Nash equilibrium in this case. Can 2 win in the Nash equilibria? [2]
 - (iii) The block votes for 1. But if 1 takes a left-wing position the block does not vote for anyone. Specifically, if 1 takes a position $x_1 < m - k$ the block does not vote. Here k is a small number and 45% of voters have their favourite positions equal or less than $m - k$. Identify any one Nash equilibrium in this case. [2]
2. [Bertrand duopoly with continuous demand functions] Consider a Bertrand oligopoly with two firms where the demand functions are given by, $q_1 = 20 - p_1 + 2p_2$ and $q_2 = 20 - p_2 + p_1$, and cost functions are given by, $C_1(q_1) = q_1$ and $C_2(q_2) = 2q_2$. Derive the best response functions of the firms and find the Nash equilibrium/equilibria. [2+2]

HS 123: Game Theory and Economics
BTech, 4th Semester, End-Semester Test
Total Marks: 40
Date: 5th May, 2024

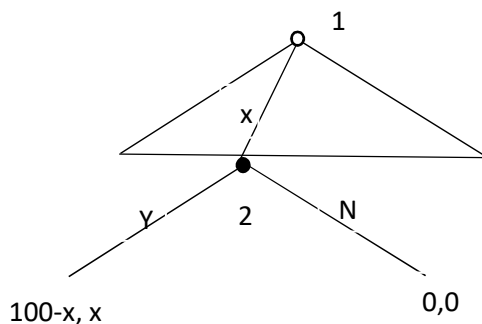
1. [Electoral competition with discrete positions, citizen candidates and cost of fighting election] Suppose there are 101 voters with fixed positions 0 to 100. The winner gets a prize of 200. Any voter can choose to contest as a candidate, but there is a cost of contesting, 100. If there is a tie between two winners, each gets 100 as the prize. If the voter with position x contests, she announces x as her policy. If voter y wins, voter x gets $-|y - x|$, because winner's policy is different from x 's position (this is in addition to the cost which x incurs in case she contests). The average of the two winners' positions becomes the policy in case of a tie. If none contests each voter gets 0. Voters decide whether to contest or not.
 - (a) Is the voter 50 contesting alone a Nash equilibrium?
 - (b) Is 49 and 51 contesting a Nash equilibrium?
 - (c) Is 11 and 89 contesting a Nash equilibrium? [3+3+4 =10]
 Explain your answer in each case.

2. Two players, players 1 and 2, are bidding for an object in a sealed bid auction. The bidding starts from 0, one can bid only in multiples of 100 rupees, the maximum bid is 500 rupees. 1 and 2 value the object 400 and 300 rupees respectively. The player making the highest bid wins the object. In case of a tie, player 1 wins the object. If it is a second price auction, find *all* Nash equilibria. [6]

3. Consider the following game with von Neumann-Morgenstern preference.
 - (a) Is there any action of any player which is strictly dominated (consider mixed strategies)? If yes, mention the mixed strategies that strictly dominate the action. [4]
 - (b) Find all Nash equilibrium of the game (both pure and mixed strategy equilibrium). What are the payoffs of the players in the equilibrium/equilibria? [4+2]

		2		
		a	B	c
1	x	2, 3	1, 1.5	4, 1
	y	4, 0	3, 2	2, 4

4. Consider the ultimatum game as discussed in the class (game tree give below), with the difference that instead of continuous, x can take only discrete values: $0, 1, 2, \dots, 100$. In these changed circumstances what is/are the subgame perfect equilibrium/equilibria of the game? Mention any one Nash equilibrium of the game. [4+2]



5. Construct the strategic form of the following extensive game and find its Nash equilibrium/equilibria. What is/are the subgame perfect equilibrium/equilibria of the game? [2+3+3]

